

Second-Hand Product Escrow Marketplace

Lab 3 – Component Modelling & Architectural Pattern Selection

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1. Architectural Choice

We chose Microservices Architecture for the Second-Hand Product Escrow Marketplace. The system is here divided into 5 main components: User & Product Management, Order Manager, Payment Service, Inspection Service, and Escrow Service. The Payment Gateway is treated as external system.

2. Scenario-Related Reasons

Separation of responsibilities	Payment, inspection, order processing, and escrow management are isolated into separate services. This prevents financial operations from being tightly coupled with inspection and marketplace functions.
Independent service operation	The Inspection Service, Payment Service, and Escrow Service can operate and be maintained independently. This fits marketplace where transaction, inspection, and payment workloads may differ.

3. Component & Interface Structure

The Order Manager coordinates the transaction and communicates through the User/Product API, Payment Processing API, Inspection Request API, and Escrow Management API. Payment Service communicates with the external Payment Gateway API, while the Inspection Service provides verification status to the Escrow Service. Provided interfaces are represented using ball notation and required interfaces using socket notation.

4. Security Advantage

The architecture isolates the Escrow Service as the component responsible for locking, holding, releasing, and freezing buyer funds. This creates a clear security boundary around escrow operations. When dispute occurs, the escrow funds can remain frozen, preventing an unrelated component from directly releasing the funds. This supports scenario requirement that funds are released only after the required verification and buyer approval.

5. Performance Benefit

Microservices allow individual services to be scaled independently according to workload. For eg, if inspection requests increase, the Inspection Service can be scaled without scaling the entire marketplace. Similarly, increased payment traffic can be handled by scaling the Payment Service independently. This can improve resource utilization and responsiveness under varying marketplace loads.

Conclusion: The component diagram represents a microservices-based marketplace in Order Manager coordinates the transaction, Payment Service handles payment processing, Inspection Service handles independent verification, and Escrow Service controls the release of buyer funds.